Branch: CSE/AI/IT/DS **Sem:- III**
Subject: NMDM/NTDM/DMSA **Subject Code: CS/AI/IT/DS3M004**
Subject In-charge : Prof. Ashvini Ramteke

QUESTION BANK

Sr. No	Questions	Unit No.	Marks
1	Define: characteristic matrix	I	2
2	Define: characteristic polynomial	I	2
3	Define: characteristic equation	I	2
4	Define: characteristic roots	I	2
5	Define: Eigen vectors	I	2
6	Interpret the term: Independent Vectors	I	2
7	Define: Dependent Vectors	I	2
8	What is consistent system ?	I	2
9	Define: System of linear equations	I	2
10	what is inconsistent system of linear equation?	I	2
11	Define : Homogeneous system of linear equation	I	2
12	write a condition when a homogeneous system has a unique solution.	I	2
13	write a condition when a non-homogenous system has an infinite solution.	I	2
14	write a condition when a homogenous system has no solution.	I	2
15	Write a statement of the Sylvester's theorem.	I	2
16	interpret the term: Quadratic form	I	2
17	interpret the term: Canonical form	I	2
18	interpret the term: Index	I	2
19	interpret the term: Rank of the matrix	I	2
20	interpret the term: Signature of quadratic form.	I	2
21	What are the types of number ?	II	2
22	Define : Approximate number	II	2
23	Define : Inherent Error	II	2
24	Define : Round off Error	II	2
25	Define :Truncation Error	II	2
26	Interpret the term: Absolute Error	II	2

27	Interpret the term: Relative Error	II	2
28	Interpret the term: Percentage Error	II	2
29	Interpret the term: Algebraic Equation	II	2
30	Interpret the term: Transcendental Equation	II	2
31	Write a formula for Bisection method	II	2
32	Write a formula for N-R method.	II	2
33	Define: Pivot	II	2
34	Write a condition for Gauss- Seidel Method.	II	2
35	Define: Mathematical expectation for DRV & CRV	III	2
36	Define: variance	III	2
37	Define: Standard Deviation	III	2
38	Write a formula for E(KX) .	III	2
39	Write a formula for Var(kX)	III	2
40	Define: Moment about a point for DRV & CRV	III	2
41	Define: Moment about an origin for DRV & CRV	III	2
42	Define: Moment about mean for DRV & CRV	III	2
43	Interpret the term: Moment generating function for DRV & CRV	III	2
44	Define : Joint Distribution of DRV	IV	2
45	Define : Joint Distribution of CRV	IV	2
46	Interpret the term: Marginal probability function of X for DRV	IV	2
47	Interpret the term: Marginal probability function of Y for DRV	IV	2
48	What id Independent Variable ?	IV	2
49	Define: Joint Distribution function.	IV	2
50	Interpret the term: Marginal probability function of X for CRV	IV	2
51	Interpret the term: Marginal probability function of Y for CRV	IV	2
52	Define: Conditional Expectation of X given Y	IV	2
53	Define: Conditional Expectation of Y given X	IV	2
54	What id binary operation ?	V	2
55	Define: Groupoid	V	2
56	What is Algebraic Structure?	V	2
57	Define: Semi-group	V	2
58	Interpret the term: Monoid	V	2
59	Write all the properties of group.	V	2
60	What will be the identity element for (R, +) & (R*, x) ?	V	2
61	What is an abelian group?	V	2
62	Define: Finite group	V	2
63	Interpret the term: Infinite group	V	2
64	What is order of a group?	V	2

65	Give two examples of a group with their respective operation.	V	2
66	Write cancellation laws.	V	2
67	What is order of an element?	V	2
68	Define: Cosets	V	2
69	State : Lagrange's theorem	V	2
70	Define: Normal group	V	2
71	Interpret the term: Homomorphism of a group	V	2
72	Interpret the term Isomorphism of a group	V	2
73	Define: Isomorphic Group	V	2
74	Define: Subgroup	V	2
75	Give an example of a group and a subgroup with their binary operation.	V	2
76	Solve the differential equation by using Matrix Method: $\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$, x(0) = 2 , x'(0) = 1	I	12
77	Reduce the diagonal form of the following symmetric matrix and interpret the result in terms of sum of squares and find the corresponding linear transformation. Also find the index and signature. $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$	I	12
78	Find the root of the equation $xsinx + cosx = 0$ with $x_0 = 3.1416$ by using Newton-Raphson Method up to forth iterations.	II	**7**
79	Find the root of the equation $x = e^{-x}$, using the Newton Raphson Method up to 4th iteration.	II	**7**
80	Apply Crout's Method to solve the following system of linear equations: $3x + 4y - z = 1$ $2x + 2y + 5z = 5$ $-x + y + 2z = 2$	II	12
81	Solve the following system of linear equation using the Gauss-Seidel Method: $4x - y + z = 7$ $-2x + 5y + 2z = -6$ $x + y + 3z = 8$	II	12
82	Illustrate the eigen values and the eigen vector for the given matrix: $\begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{bmatrix}$	I	12
83	Illustrate the eigen values and eigen vectors for the given matrix: $\begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}$	I	12

<table>
<tr><td>84</td><td>Test for the consistency and solve the following system of linear equations :
$$2x + 3y - z = 5$$
$$4x - y + 2z = 3$$
$$-6x + 9y + 3z = 7$$</td><td>I</td><td>7</td></tr>
<tr><td>85</td><td>Inspect whether the vectors $X_1 = [1,2,3], X_2 = [4,5,6]$, $X_3 = [7,8,9]$, $X_4 = [10,11,12]$ are linearly dependent or not. Also find their linear combination.</td><td>I</td><td>7</td></tr>
<tr><td>86</td><td>Solve the differential equation by using Matrix Method:
$\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$, x(0) = 3 , x’(0) = 2</td><td>I</td><td>12</td></tr>
<tr><td>87</td><td>Reduce the diagonal form of the following symmetric matrix and interpret the result in terms of sum of squares and find the corresponding linear transformation. Also find the index and signature.
$$\begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$</td><td>I</td><td>12</td></tr>
<tr><td>88</td><td>Find the root of the equation $-1.25x^5 + 2.5x^4 - 3x + 6 = 0$ by using Newton-Raphson Method up to forth iterations.</td><td>II</td><td>7</td></tr>
<tr><td>89</td><td>Find the root of the equation $x = e^{-2x}$, using the Newton Raphson Method up to 4th iteration.</td><td>II</td><td>7</td></tr>
<tr><td>90</td><td>Construct the Cayley Table for the set G={1,2,3,4,5,6,7,8} with the operation multiplication modulo 9 and check whether the given set forms an abelian group or not . Also find the order of each element if the set is a group.</td><td>V</td><td>7</td></tr>
<tr><td>91</td><td>Let X be a random variable with probability distribution as
<table>
<tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr>
<tr><td>P(X)</td><td>1/8</td><td>3/8</td><td>3/8</td><td>1/8</td></tr>
</table>
Inspect the following terms:
E(X) 2)E(X^2) 3) E($2X^2 + 7$) 4) E(X^3) 5) Var(X)</td><td>III</td><td>12</td></tr>
<tr><td>92</td><td>Let X and Y be random variables having probability function as
$$f(x) = \begin{cases} c\,(x^3 + y^3) & , x = 0,1,2\ ;\ \ y = 0,1,2 \\ 0 & , \quad otherwise \end{cases}$$
Inspect the following terms:
1) C
2) Marginal Densities
3) Find the conditional expectation of Y given X and X given Y.</td><td>IV</td><td>12</td></tr>
<tr><td>93</td><td>Let X and Y be continuous random variables having joint density function:
$$f(x) = \begin{cases} c(3x + 1)\ , & 0 < x < 2\ ;\ \ 0 < y < 3 \\ 0\ , & otherwise \end{cases}$$
Inspect the following terms:
1) C</td><td>III</td><td>12</td></tr>
</table>

	2) The marginal density function of X & Y 3) P(X > ½ , Y < 3/2)		
94	A random variable X has density function given by $f(x) = \begin{cases} 3e^{-3x} & ,x \geq 0 \\ 0, & x < 0 \end{cases}$ Solve to Find the moment generating function and first four moments about origin. Also find the E(X) and Var(X).	III	7
95	Use the Newton -Raphson Method to find the root of the equation $f(x) = x^3 - 2x - 5 = 0$ starting with an initial guess of $x_0 = 2$ up to 3 iterations .	II	7
96	Find the root of the equation $f(x) = e^x - 3x = 0$ using the N-R method . Start with the initial guess $x_0 = 1$ up to 2 iterations.	II	4
97	Using the N-R method to find the root of the equation $f(x) = x^4 - 2x^2 - 3 = 0$ starting with $x_0 = 1.5$ up to 2 iterations	II	4
98	Apply Bisection method to find the roots of the function $f(x) = cosx - x = 0$ within the interval [0,1]. Perform 4 iterations	II	7
99	Apply Bisection method to find the roots of the function $f(x) = x^3 + x - 4 = 0$ within the interval [1,2]. Perform 4 iterations	II	4
100	Apply Bisection method to find the roots of the function $f(x) = x^2 - 2x - 3 = 0$ within the interval [-2, -1]. Perform 4 iterations	II	7
101	Use the Newton -Raphson Method to find the root of the equation $f(x) = x^3 + x - 4 = 0$ in the interval [1,2] up to 3 iterations .	II	7
102	Find the root of the equation $f(x) = e^x - 3x = 0$ using the N-R method . Start with the initial guess $x_0 = 1$ up to 2 iterations.	II	4
103	Using the N-R method to find the root of the equation $f(x) = x^5 - 2x^2 - 3 = 0$ up to 2 iterations	II	4
104	Apply Bisection method to find the roots of the function $f(x) = cosx - 5sinx = 0$ within the interval [0,1]. Perform 4 iterations	II	7
105	Apply Bisection method to find the roots of the function $f(x) = x^3 + x - 4 = 0$ within the interval [1,2.5]. Perform 4 iterations	II	4
106	Apply Bisection method to find the roots of the function $f(x) = x^2 - x - 3 = 0$ within the interval [-2, -1]. Perform 4 iterations	II	7
107	Apply Crout's Method to solve the following system of linear equations: $x - y + 2z = 5$ $3x + y - z = 3$ $2x + 3y + z = 10$	II	12
108	Solve the following system of linear equation using the Gauss-Seidel Method up to 2 iterations: $x - y + 2z = 5$ $3x + y - z = 3$ $2x + 3y + z = 10$	II	7

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111	Illustrate the eigen values and the eigen vector for the given matrix: $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$	I	12
112	Illustrate the eigen values and eigen vectors for the given matrix: $\begin{bmatrix} 6 & 2 & 1 \\ 2 & 6 & 2 \\ 1 & 2 & 6 \end{bmatrix}$	I	12
113	Illustrate the eigen values and the eigen vectors of the matrix: $A=\begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$	I	7
114	Illustrate the eigen values and the eigen vectors of the matrix: $A=\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$	I	7
115	Given system of linear equations represented by the augmented matrix: $\left[\begin{array}{ccc\|c} 1 & 2 & 3 & 9 \\ 2 & 4 & 6 & 18 \\ 3 & 6 & 9 & 27 \end{array}\right]$ Check if the system is consistent , inconsistent or has infinitely many solutions.	I	12
116	Test for the consistency and solve the following system of linear equations : $$x-y+2z=3$$ $$2x-2y+4z=6$$ $$3x-3y+6z=10$$	I	12
117	Given system of linear equations represented by the augmented matrix: $\left[\begin{array}{ccc\|c} 2 & 1 & 3 & 7 \\ 4 & 2 & 6 & 10 \\ 1 & 2 & 1 & 5 \end{array}\right]$ Check if the system is consistent , and find it's solutions if possible.	I	12
117	Test for the consistency and solve the following system of linear equations : $$3x+2y-z=1$$ $$6x+4y-2z=2$$ $$x+y+z=3$$	I	12

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$$f(x) = \begin{cases} c\,(x^2 + y^2) & , x = 0,1,2\,; \quad y = 0,1,2 \\ 0 & , \quad otherwise \end{cases}$$
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$$f(x) = \begin{cases} 3c\,(x^2 + y^2) & , x = 0,1,2\,; \quad y = 0,1,2 \\ 0 & , \quad otherwise \end{cases}$$
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4) C
5) The marginal density function of X & Y
6) P(X > ½ , Y < 3/2)</td><td>IV</td><td>12</td></tr>
<tr><td>122</td><td>Let G=R-{0} the set of non-zero real numbers with the operation of multiplication. Determine whether G is a group under multiplication.</td><td>V</td><td>7</td></tr>
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<tr><td>124</td><td>Let G={0,1,2,3,4,5} under addition modulo 6. Determine if G is a group under addition modulo 6.</td><td>V</td><td>7</td></tr>
<tr><td>125</td><td>Let G={0,1,2,3,4} forms a group under addition modulo 5. Determine the order of each element.</td><td>V</td><td>4</td></tr>
<tr><td>126</td><td>Let G={0,1,2,3,4,5} under addition modulo 6. Determine the order of each element.</td><td>V</td><td>4</td></tr>
</table>

127	Consider the group G={0,1,2,3,4,5} under addition modulo 6. Check if the set H={0,3} is a subgroup of G.	V	7
128	Consider the group ($\mathbb{Z}$,+) the set of all integers under addition. Check if the set H=3$\mathbb{Z}$ ={……,-6,-3,0,3,6……} forms a subgroup of G.	V	7
129	Let G={0,1,2,3,4,5} under addition modulo 6 and H={0,3} . Find the left and right cosets of H in G	V	7
130	Let G = $\mathbb{R}$-{0} be a group of non-zero real numbers with multiplication operation and let H={1,-1} , a subgroup of G. Find the left and right cosets of H in G.	V	7

Subject In-charge **Academic In-charge** **Dept. HOD**